

ALENIGLIPRON vs ELECOGLIPRON — PHASE 2

MATHEMATICAL MODEL + THE LIVER POST-MORTEM

2026-09-04 · Desk analysis (FABLE). Inputs: ACCESS/ACCESS-II 8-Ks + Nature Medicine table, AZ H1-26 6-K (VISTA/SOLSTICE), FDA labels, gap-sweep primaries. Items sourced from stable pre-2026 knowledge rather than this session's pulls are tagged [K].

PART A — What actually killed the small-molecule livers

Five oral small-molecule GLP-1 agonists now have meaningful human exposure. The scoreboard:

Molecule	Max daily dose	Liver outcome
Orforglipron (Lilly)	17.2mg commercial / 36mg trial	CLEAN — approved label has zero hepatic warnings
Elecglipton (AZ)	75mg (top Ph2)	UNDISCLOSED (efficacy-only release; 24k-pt Ph3 = indirect confidence)
Aleniglipron (Structure)	180mg Ph3 / 240mg tested	No DILI at n>625; 8 transient $\geq 3 \times \text{ULN}$, 1 $\geq 5 \times$, 0 $\geq 10 \times$ ($\approx 1.3\%$ excursion rate, all resolving ON drug)
Danuglipron (Pfizer)	$\sim 400\text{mg/day}$ (200mg BID) [K]	KILLED Apr-2025: one asymptomatic potential DILI, resolved; Pfizer cited “totality... input from regulators”
Lotiglipron (Pfizer)	— [K]	KILLED Jun-2023 on transaminase elevations [K]
TERN-601 (Terns)	$\sim 740\text{mg/day}$ at top [K-approx]	KILLED Oct-2025: 3 grade-3 elevations, 2 adjudicated DILI (1 biopsy-confirmed)

The mechanism is chemistry, not pharmacology. Three exonerating facts: GLP-1R is not meaningfully expressed on hepatocytes; the peptide agonists (millions of patient-years) show no hepatotoxicity — the class *improves* liver fat (semaglutide’s MASH program); and the failures don’t share a scaffold with the survivors. What the failures DO share is **daily-dose burden**: the liver clears these molecules through oxidative metabolism, and idiosyncratic DILI risk scales with the mass of drug metabolized daily (the classic $>100\text{mg/day}$ heuristic [K] — reactive-metabolite load is roughly proportional to dose). The failures sit at $\sim 400\text{--}740\text{mg/day}$; the clean approvals at 17–36mg. Danuglipron’s BID regimen additionally doubled the peak-exposure cycles per day. So: **molecule-specific in expression, dose-burden-mediated in mechanism** — the class is acquitted, but high-milligram members carry structural risk.

Where that leaves aleniglipron: in the middle, with a bounded but real residual. 180–240mg sits 5–10x above orforglipron and 2.4–3.2x above elecglipton. The clean n>625 record bounds true DILI incidence below **0.48%** (rule of 3, 95%) — which still permits **up to ~17 cases in the**

3,600-patient ACCOMPLISH-1 consistent with everything seen so far. Note the asymmetry in what killed danuglipron: ONE asymptomatic resolved case in >1,400 was enough for Pfizer post-lotiglipron. Aleniglipron already has 8 transient excursions on the record; the difference between “transient excursion” and “DILI” is persistence and causality adjudication — at Phase 3 scale, with 10x orforglipron’s dose burden, this is the honest tail risk the clean headline doesn’t extinguish. It is also why the FDA-label contrast matters so much: orforglipron earned a §5 with *nothing* hepatic in it; that is the bar.

PART B — The Phase 2 mathematics, five results

B1. The 0.2pp “gap” is 0.1σ — unresolvable, permanently, at Phase 2 scale. Aleniglipron 120mg: -11.3 (CI $-13.9, -8.6 \rightarrow SE \approx 1.35pp$). Elecoglipron 75mg: -11.5 (no CI disclosed; similar N $\rightarrow$ similar SE). SE of the difference $\approx 1.9pp$; the observed 0.2pp difference is **0.10σ** . Detecting a true 2pp difference at 80% power needs $\sim 250/arm$ (these ran 56–65); detecting 1pp needs $\sim 1,000/arm$. **No Phase 2 in this class can rank these molecules. Anyone claiming an efficacy ordering from these trials is reading noise.**

B2. The weight-gaining placebo is a $\sim 7\%$ event — which is exactly why you don’t headline it. P(a ~ 20 -patient placebo arm drifts to $+1.1\%$ when truth is $\approx -0.7\%$, $\sigma \approx 5.5\%$) $\approx 7.2\%$. Not aberrant — a one-in-fourteen draw. But the company’s headline (-16.3 adjusted) inherits the full $+1.9pp$ of placebo divergence. Small-N placebo arms are lottery tickets; ACCESS-II drew a flattering one.

B3. Trajectory fitting kills the “no plateau” story as marketed. Fit $W(t) = W^\infty(1 - e^{-(t/\tau)})$: - **Elecoglipron** (26wk 10.5% $\rightarrow$ 36wk 11.8% absolute, same study): **$\tau \approx 17wk$, $W^\infty \approx 13.4\%$ absolute** — predicted 76-week value $\approx 13.3\%$; the curve is $\sim 90\%$ saturated already. - **Aleniglipron** cannot be fit within one study (one timepoint). Forcing a fit through the 56-week open-label 16.2% requires **$\tau \approx 60wk$ and $W^\infty \approx 27\%$ absolute** — an asymptote no oral or injectable mono-agonist has ever approached. The far likelier reading: the OLE point is inflated by **completer/survivor bias** (dropouts lose less and exit the denominator) plus open-label effects. Correcting to plausible τ (25–35wk), aleniglipron’s true asymptote is **$\sim 15\text{--}17\%$ absolute** — better than elecoglipron’s ~ 13.4 if the within-study points are taken at face value, but the comparison is τ -model-dependent and elecoglipron has no $>36wk$ data. **“No plateau” is what an exponential looks like at $t < 2\tau$ — both companies say it; neither curve supports a claim beyond ordinary saturation kinetics.**

B4. Two independent methods put aleniglipron-180’s true effect at $\approx -12.4pp$, not -16.3 . (a) *Dose–response*: ACCESS’s own within-study curve ($-8.2/-9.8/-11.3$ at 45/90/120mg) runs $\sim 2pp$ per dose-doubling; extrapolating the 0.58 doublings to 180mg $\rightarrow$ **$-12.5pp$** . (b) *Bayesian shrinkage*: prior = the oral class (four independent programs at $-11.3/-11.5/-11.6/-11.8$, class $\sigma \approx 1.0$); likelihood = ACCESS-II’s -16.3 with $SE \approx 2.0$ ($n \approx 20/arm$) $\rightarrow$ posterior **$-12.4pp$ (SE 0.9)**. Two methods, no shared assumptions, same answer. **The desk’s estimate of aleniglipron-180’s true placebo-adjusted effect: $-12.4 \pm 1.0pp$ — modestly above class, nowhere near the marketed 16.3.**

B5. FROZEN PREDICTION — ACCOMPLISH-1, registered for grading (2028). Start from -12.4 (B4), add $\sim +1.5$ pp for 36→76-week duration (decelerating curve), subtract the estimand haircut (on-treatment→regulatory ITT: orforglipron’s own registry-vs-label gap was 2.5pp at 10–24% missingness; assume 1.5–2.8pp at aleniglipron’s expected discontinuation): **predicted ACCOMPLISH-1 placebo-adjusted ITT at week 76: -11.1 to -12.4 pp, point -11.7 , conditional on no hepatic event** (Part A tail). Versus Foundayo’s label -9.0 : a real but modest ~ 2.5 pp edge — enough to matter clinically, nowhere near the 4–7pp the -16.3 framing implies. If the print lands in-band, the estimand model is validated; below -10 , the bear owns the name; above -13.5 , we were wrong in the good direction and the shrinkage prior was too strong.

Near-term corollary (Q3 pack, gradeable this month): expect the OLE 72-week release to headline **$\geq 16.5\%$ absolute** (open-label completer arithmetic makes this near-mechanical, $p \approx 0.7$) — treat that number as NON-decisive regardless of size; the decisive lines remain mature discontinuation and any hepatic table.

Bottom line on “which is better”

The molecules are efficacy-tied beyond Phase 2’s power to resolve (B1); aleniglipron’s headline premium is placebo-draw + estimand + small-N artifacts (B2–B4) over a true edge of perhaps ~ 1 pp at matched dose-response — bought at 2.4x elecoglipron’s dose burden, which is precisely the axis the class’s failures died on (Part A). Elecoglipron’s missing tolerability table and aleniglipron’s OLE discontinuation print are the two facts that will actually rank them.

ADDENDUM (2026-09-04): ELECOGLIPRON’S PHASE 3 UNDER THE SAME MODEL + DIRECT COMPARISON

B6. FROZEN PREDICTION — EMBOLD (elecoglipron Phase 3)

Same machinery, cleaner inputs ($N=310$, ~ 60 /arm, well-behaved placebo means the shrinkage barely moves it): elecoglipron-75’s posterior true effect is **-11.44 (SE 0.80)** — it sits exactly at the class mean with better precision than aleniglipron’s estimate. But its trajectory is $\sim 90\%$ saturated ($\tau \approx 17$ wk, $W_\infty \approx 13.4\%$ absolute $\rightarrow 13.2\%$ by week 72; only $+0.4$ pp of duration uplift left), so after the same 1.5–2.8pp ITT estimand haircut:

EMBOLD predicted placebo-adjusted ITT at week 72–76: -9.0 to -10.3 , point -9.7 — essentially a replication of Foundayo’s label (-9.0) — CONDITIONAL on 75mg being the top Phase 3 dose. The conditional matters: Embold’s dose arms are not yet disclosed, and AZ has its own dose-response curve to climb. At a 100mg top dose the point moves to ~ -10.5 ; at 150mg, ~ -11.7 — identical to our ACCOMPLISH prediction. **The entire modeled Phase 3 gap between these drugs (~ 2.0 pp in aleniglipron’s favor) is a DOSE choice, not a molecule difference** — aleniglipron “wins” the model only because Structure is running 180mg while AZ disclosed 75mg, and dose is exactly the axis the class’s liver failures died on (Part A).

Registered alongside the ACCOMPLISH prediction; grading a competitor’s print is free calibration.

THE DIRECT COMPARISON

Trial mathematics:

	Aleniglipron (Structure)	Elecglipton (AZ)
Powered Ph2 result (adj)	-11.3 @36wk, 120mg (N=230)	-11.5 @36wk, 75mg (N=310)
Observed gap	0.2pp = 0.10σ — pure noise	
Marketed headline	-16.3 (N=85, safety-primary, +1.1% placebo)	none marketed beyond the 6-K
Headline honesty	inflated: 7%-probability placebo draw + estimand + n≈20 arms	conservative: single disclosure, clean placebo, unknown estimand
Desk true-effect estimate	-12.4 ± 1.0 (at 180mg)	-11.4 ± 0.8 (at 75mg)
Precision of that estimate	worse (small arms, cross-study)	better (one clean study)
Trajectory	asymptote ~15–17% abs (OLE 16.2 is survivor-inflated; implied 27% rejected)	asymptote ~13.4% abs, ~90% saturated
Dose-response headroom	at 180mg, near its own curve's top	undisclosed above 75mg — real upside optionality
Ph3 prediction (frozen)	-11.1 to -12.4 ITT @76wk	-9.0 to -10.3 @75mg (-11.7 if 150mg)
Tolerability disclosed	fully: nausea 65–71%, vomit 32–45%, disc 10.4% (5mg start) / 2–3.4% interim (2.5mg)	NOTHING — the black box
Ph3 scale	4,700 pts, 2 trials, obesity+T2D	24,301 pts, 9 trials, incl. HF + CKD outcomes

The molecules (tagged: much of the chemistry is undisclosed on both sides):

	Aleniglipron	Elecglipton
Daily dose burden	180–240 mg — the model's central risk	75 mg (2.4–3.2x lower)
Liver implication (Part A)	middle of the danger axis; rule-of-3 permits ~17 Ph3 cases	lower burden = structurally lower idiosyncratic-DILI prior; but zero disclosed hepatic data
Regimen	once daily, 2.5mg start, 4-week titration steps	once daily [AZ-stated]; titration undisclosed
Food effect	never disclosed (the cheap open question)	undisclosed
Bias/pharmacology claim	"G-protein-biased" — COMPANY-ASSERTED, zero PubMed	partial/biased agonist per AZ framing — likewise undisclosed
GI profile as kinetics proxy	nausea 65–71% at efficacious doses — no evidence the bias decoupled aversion	unknown; SOLSTICE held a semaglutide head-to-head AZ hasn't shown
Origin / royalty burden	in-house (Gasherbrum); royalty-free , 24 patent families to 2041–45	licensed from Eccogene (China) — economics undisclosed
Owner's balance sheet	\$1.34B, dilution before data is base case	AZ — no constraint; dapagliflozin combo franchise attached

Reading it straight: as *molecules*, the honest statement is that they are efficacy- equivalent within measurement power, elecoglipron achieves the class-standard effect at a third of the metabolic load (the single most important molecular fact on the table given how this class fails), and aleniglipron's only demonstrated molecular edge — more effect at more dose — is the expensive kind. As *data packages*, aleniglipron is far more transparent (full tolerability, published hepatic tables, peer review) while elecoglipron is an efficacy press release wearing a 24,301-patient vote of confidence. As *bets*, they aren't comparable at all: one is a \$2.1B single-asset equity that must dilute, the other is a rounding error inside AZ. The two disclosures that re-rank everything: Embold's dose arms (posts to the registry eventually — watch it), and any elecoglipron tolerability table.